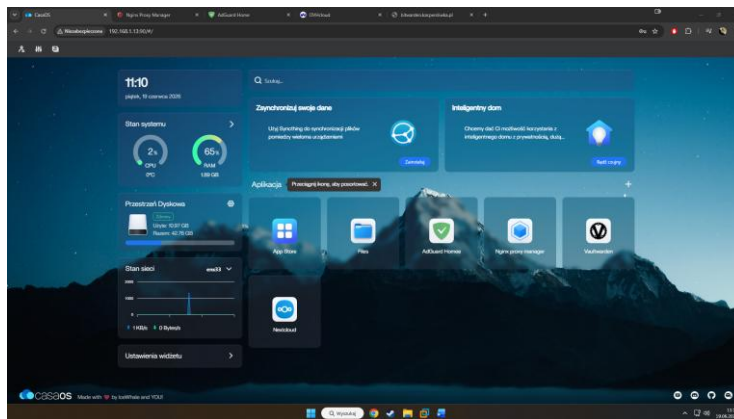
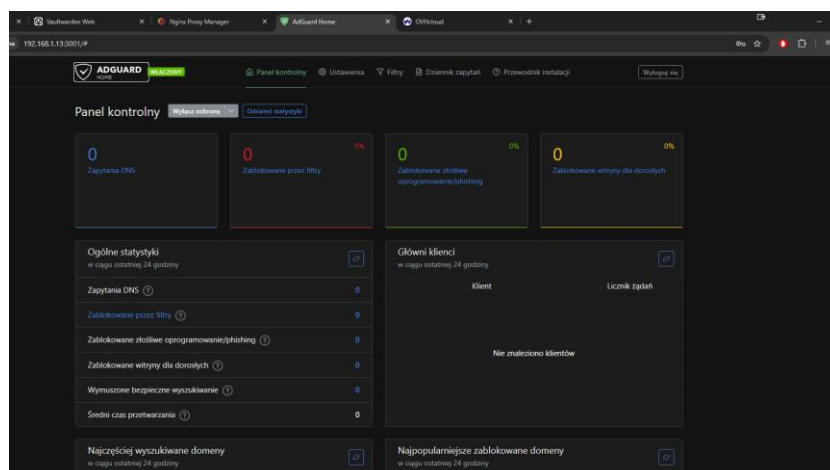
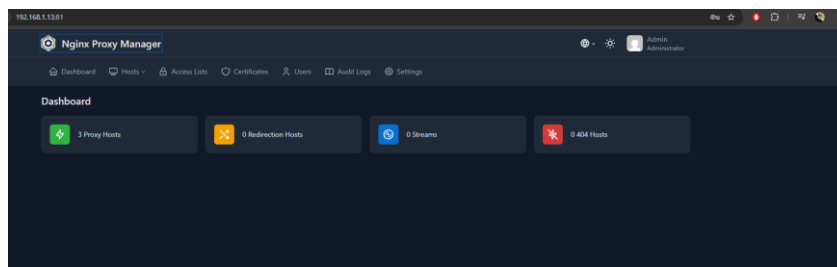
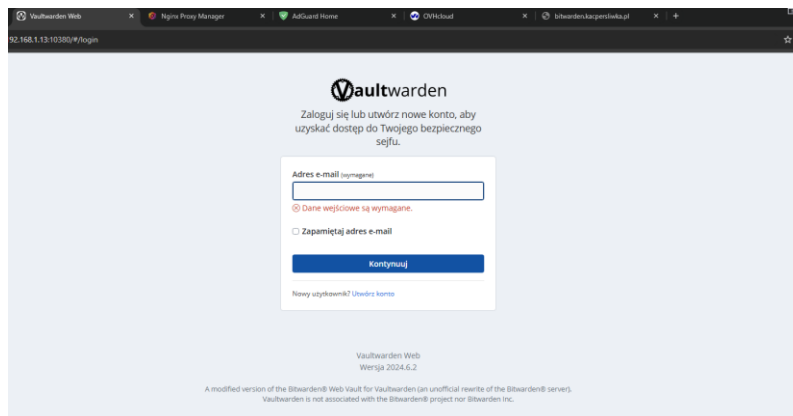


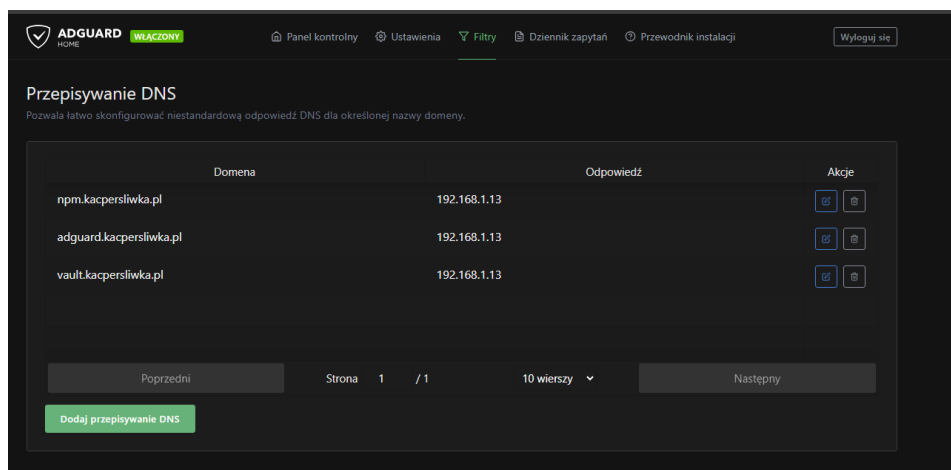
Zad 1.



Widok głównego panelu zarządzania CasaOS z zainstalowanym środowiskiem Docker oraz zestawem podstawowych kontenerów.



Uruchomione 3 serwisy



```
root@lets:~# certbot certonly --manual --preferred-challenges dns -d *.kacpersliwka.pl --key-type=ecdsa --elliptic-curve
--secp256r1
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for *.kacpersliwka.pl

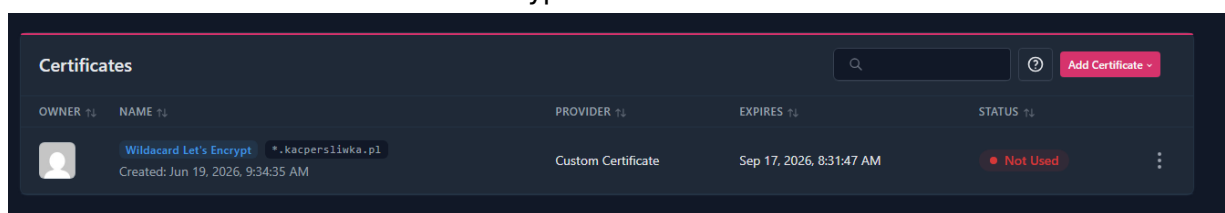
-----
Please deploy a DNS TXT record under the name:
_acme-challenge.kacpersliwka.pl.
with the following value:
20QVRptxD2mVbFtLyLiDJBdHuleOe3lHMG_TvET7U
Before continuing, verify the TXT record has been deployed. Depending on the DNS
provider, this may take some time, from a few seconds to multiple minutes. You can
check if it has finished deploying with aid of online tools, such as the Google
Admin Toolbox: https://toolbox.googleapps.com/apps/dig/#/TXT/_acme-challenge.kacpersliwka.pl.
Look for one or more bolded line(s) below the line 'ANSWER'. It should show the
value(s) you've just added.
-----
Press Enter to Continue

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/kacpersliwka.pl-0002/fullchain.pem
Key is saved at: /etc/letsencrypt/live/kacpersliwka.pl-0002/privkey.pem
This certificate expires on 2026-09-17.
These files will be updated when the certificate renews.

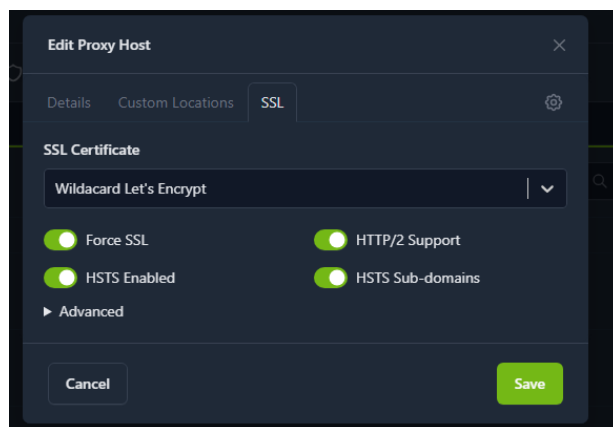
NEXT STEPS:
- This certificate will not be renewed automatically. Autorenewal of --manual certificates requires the use of an authentication hook script (--manual-auth-hook) but one was not provided. To renew this certi
cate, repeat this same certbot command before the certificate's expiry date.

-----
If you like Certbot, please consider supporting our work by:
* Donating to EFF: https://letsencrypt.org/donate
* Donating to EFF: https://eff.org/donate-le
-----
root@lets:~#
```

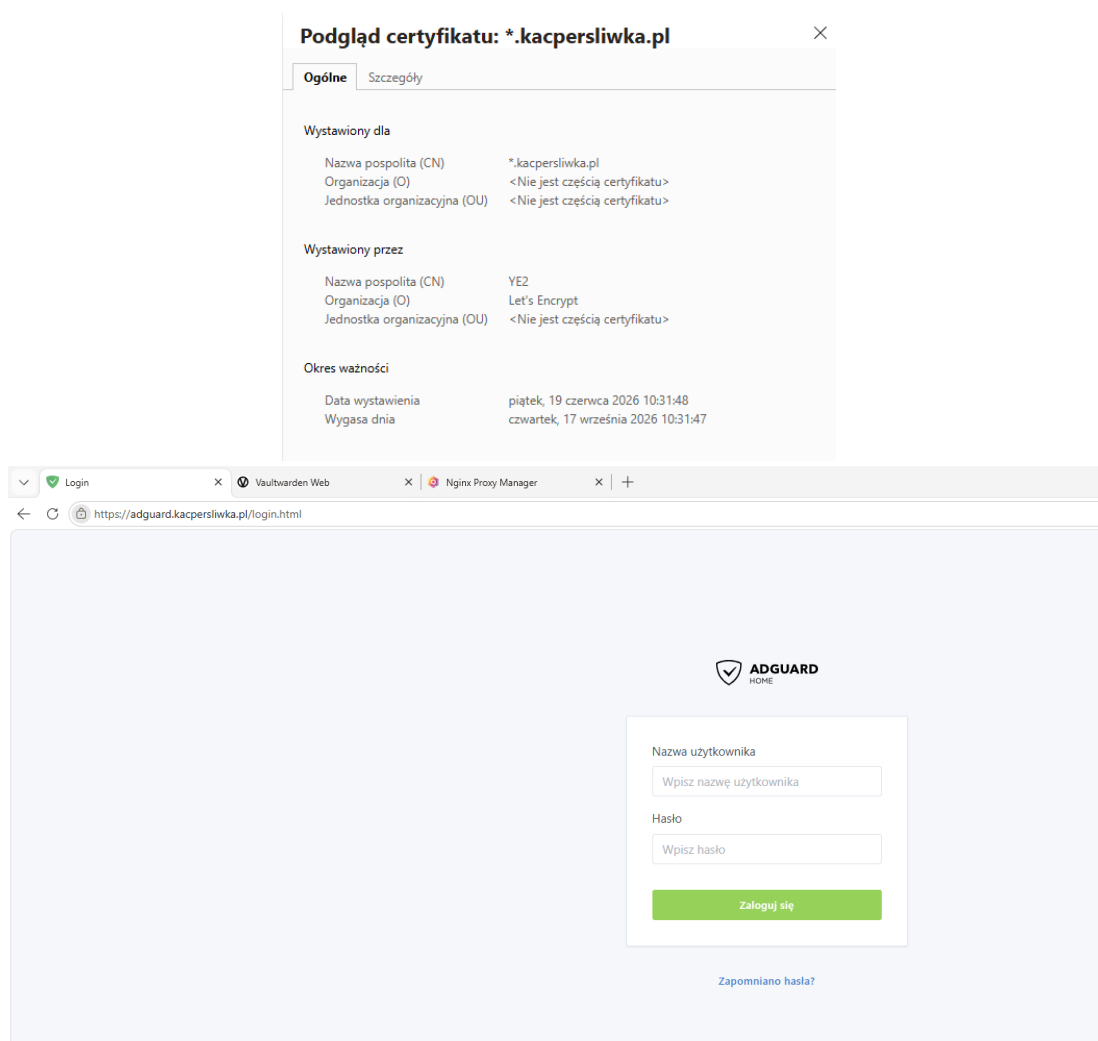
Pomyślnie wygenerowany i zweryfikowany certyfikat urzędu certyfikacji Let's Encrypt typu Wildcard

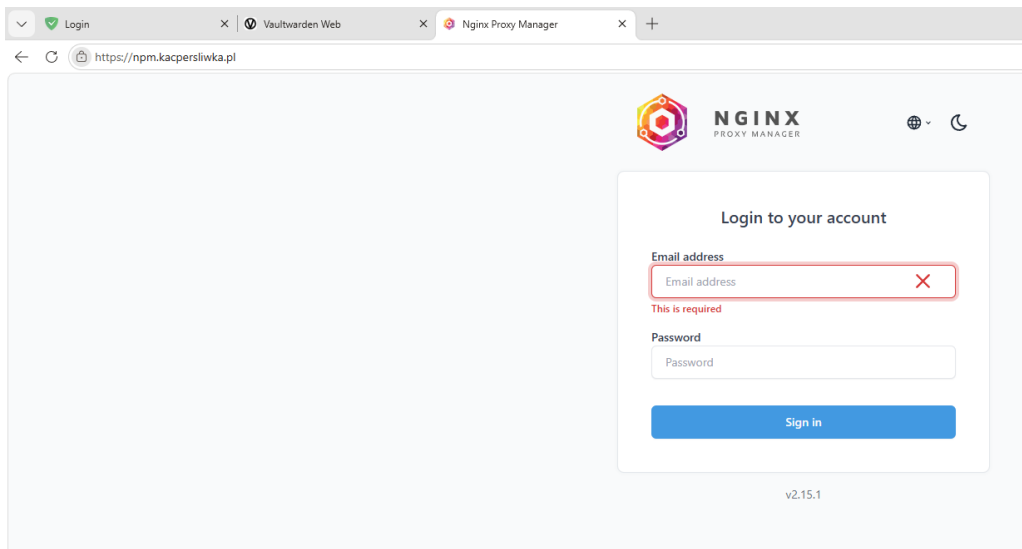
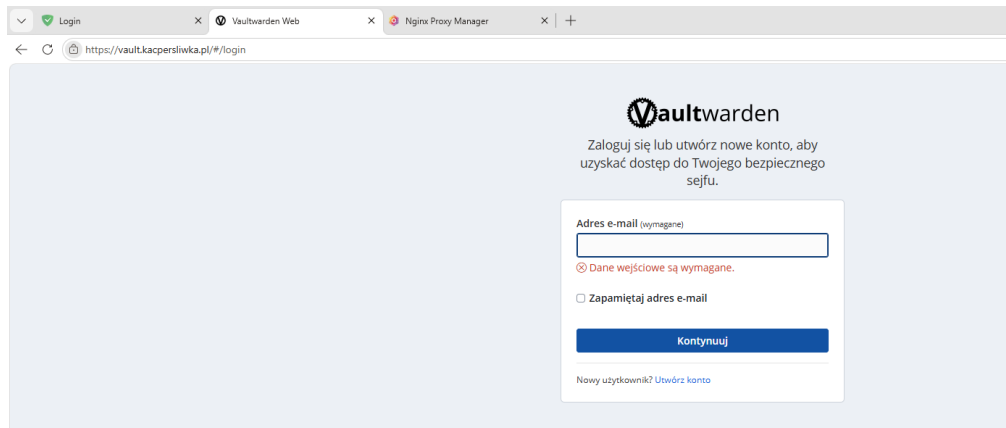


Import kluczy certyfikatu Wildcard do reverse proxy w celu automatyzacji szyfrowania poddomen.



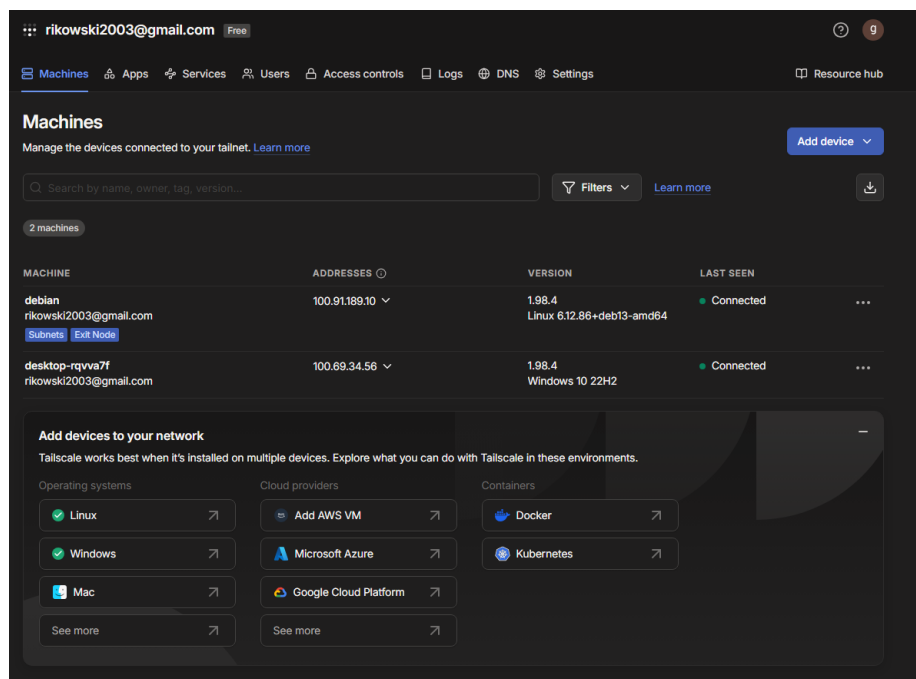
Po dopisaniu certyfikatu do proxy hostów, każda nasza strona jest zabezpieczona co widać poniżej.



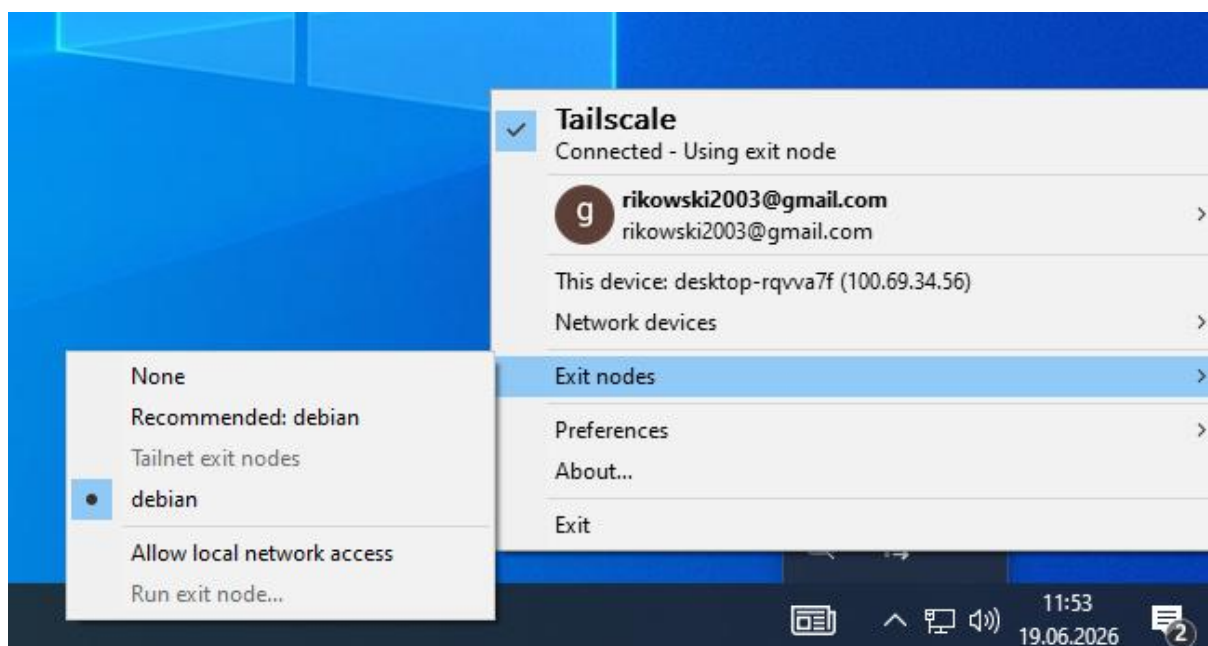


```
debian@debian: ~  
Need to get 37.1 MB of archives.  
After this operation, 73.0 MB of additional disk space will be used.  
Get:1 https://pkgs.tailscale.com/stable/debian trixie/main amd64 tailscale amd64  
1.98.4 [37.1 MB]  
Get:2 https://pkgs.tailscale.com/stable/debian trixie/main all tailscale-archive-  
keyring all 1.35.181 [3,082 B]  
Fetched 37.1 MB in 1s (60.1 MB/s)  
Selecting previously unselected package tailscale.  
(Reading database ... 139509 files and directories currently installed.)  
Preparing to unpack .../tailscale_1.98.4_amd64.deb ...  
Unpacking tailscale (1.98.4) ...  
Selecting previously unselected package tailscale-archive-keyring.  
Preparing to unpack .../tailscale-archive-keyring_1.35.181_all.deb ...  
Unpacking tailscale-archive-keyring (1.35.181) ...  
Setting up tailscale-archive-keyring (1.35.181) ...  
Setting up tailscale (1.98.4) ...  
Created symlink '/etc/systemd/system/multi-user.target.wants/tailscaled.service'  
→ '/usr/lib/systemd/system/tailscaled.service'.  
+ [ false = true ]  
+ set +x  
Installation complete! Log in to start using Tailscale by running:  
  
tailscale up  
root@debian:~#
```

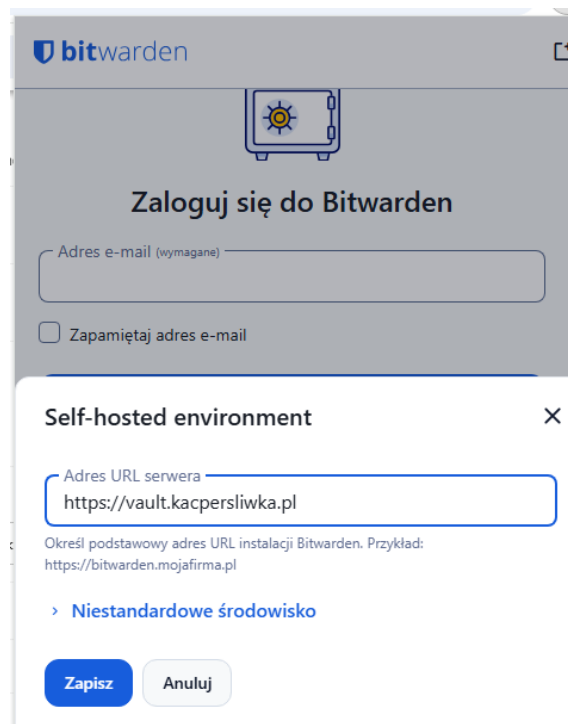
Instalacja tailscale.



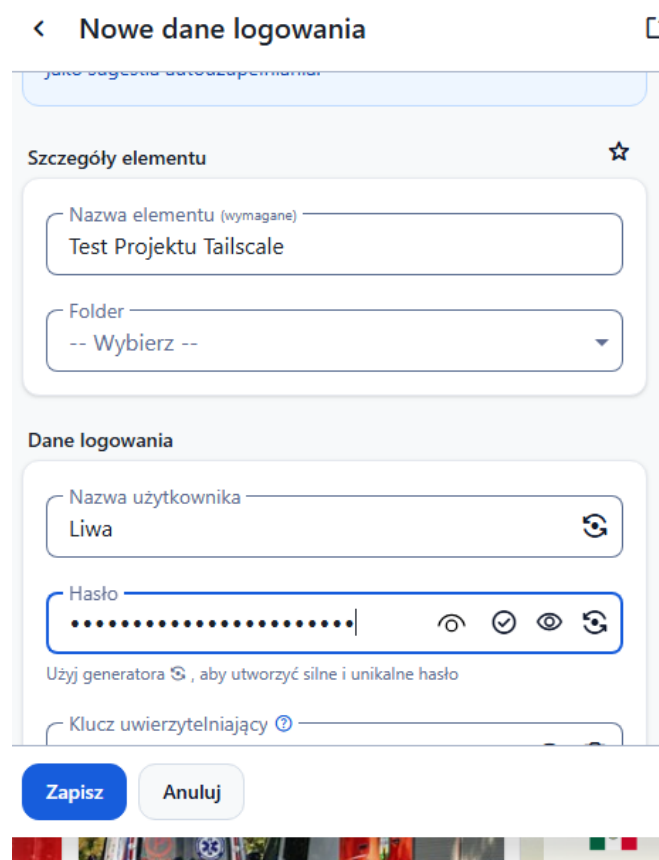
Dodanie maszyn do tailscale, serwera debian oraz systemu Windows.



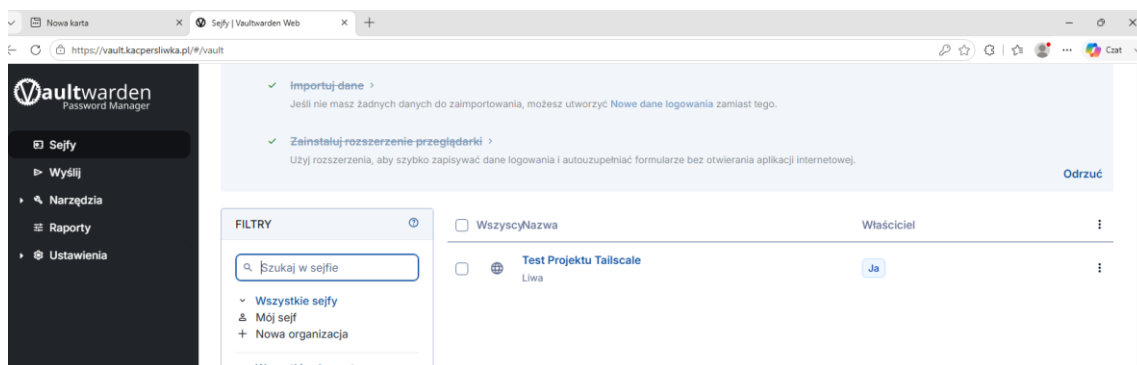
Dodanie exit node jako mój serwer, na którym jest postawiony tailscale oraz CasaOS.



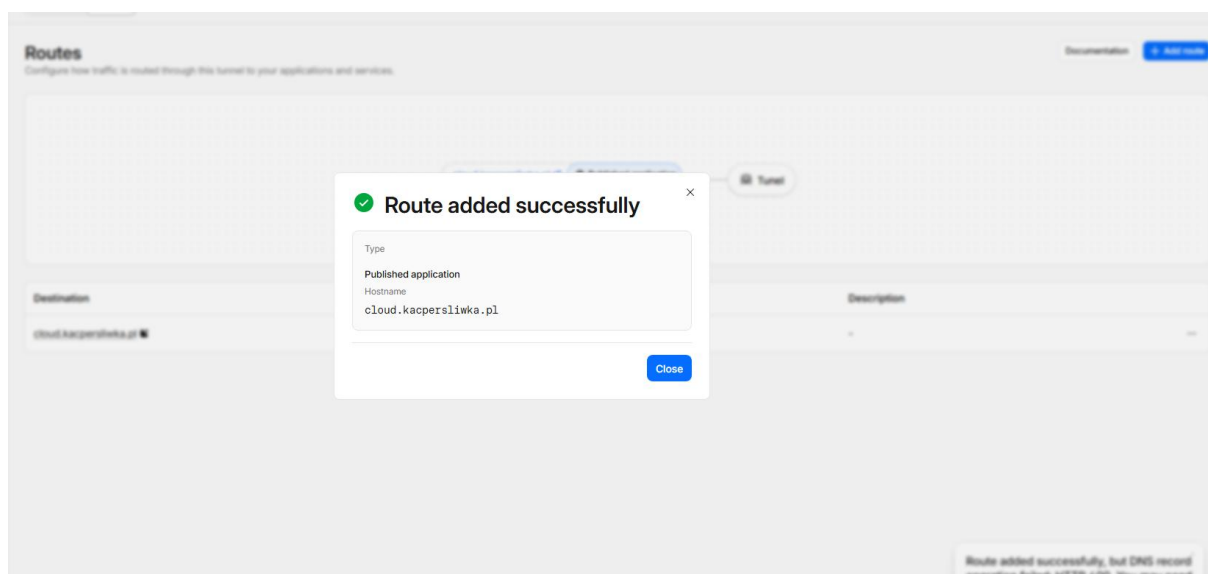
Wpisanie adresu mojego prywatnego serwera w ustawieniach wtyczki Bitwarden.



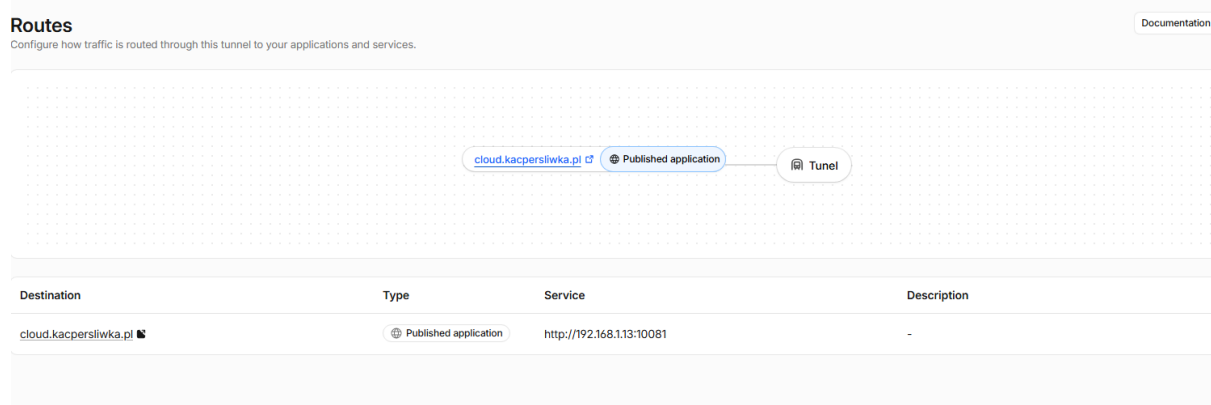
Tworzenie testowego wpisu z hasłem bezpośrednio w rozszerzeniu przeglądarki.



Weryfikacja synchronizacji – hasło dodane we wtyczce od razu pojawiło się w panelu przez stronę WWW.



Konfiguracja tunelu CloudFlare – podpięcie domeny i wskazanie lokalnego adresu IP oraz portu Nextcloud (192.168.1.13:10081).



Prawidłowo uruchomiony i działający ekran logowania do naszej chmury Nextcloud.

```
C:\Users\Admin>nslookup cloud.kacpersliwka.pl
Server:
Address: 192.168.1.1

Non-authoritative answer:
Name:    cloud.kacpersliwka.pl
Address: 192.168.1.13

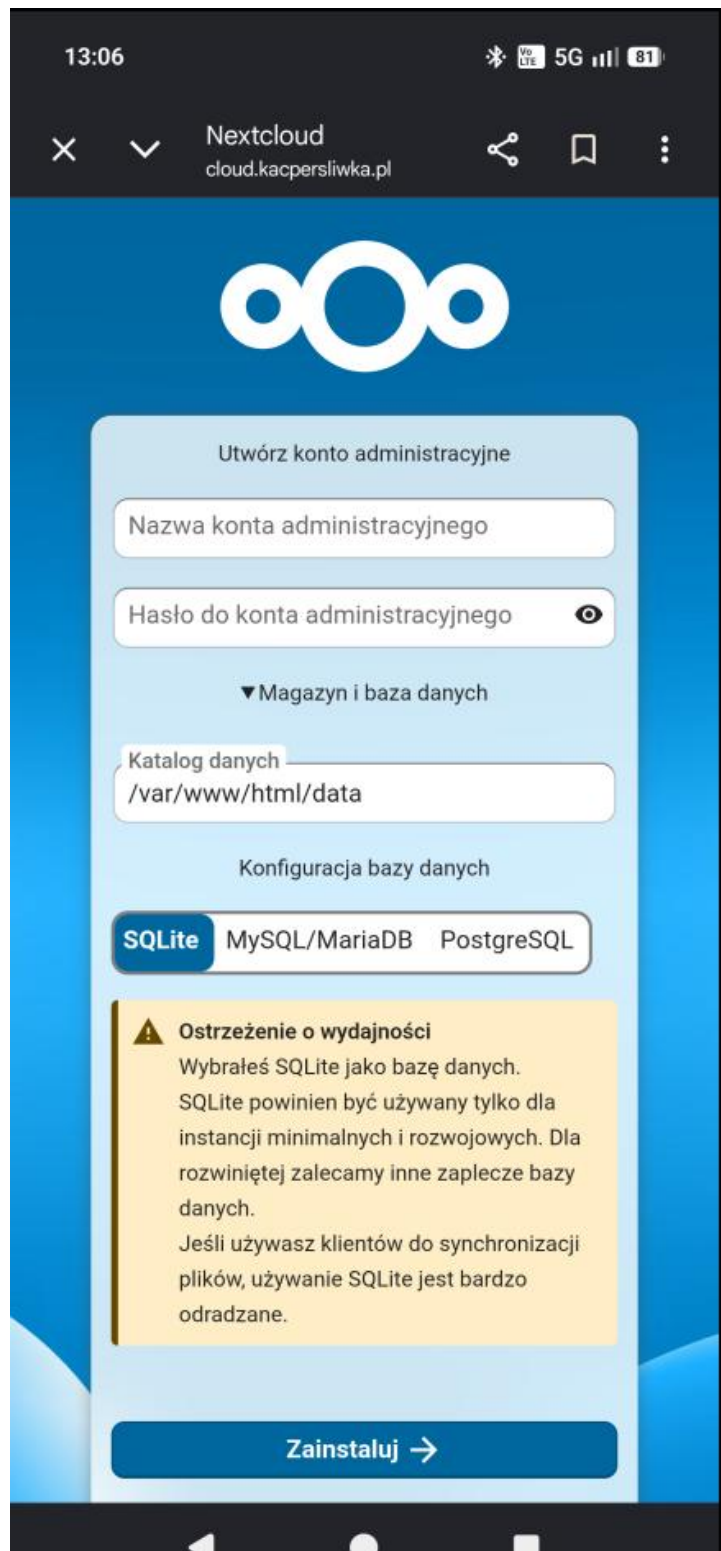
C:\Users\Admin>
```

Działanie w sieci domowej (LAN) – dzięki AdGuard Home łączę się z chmurą bezpośrednio po kablu/Wi-Fi, omijając Internet.

```
C:\Users\CL1>nslookup cloud.kacpersliwka.pl
Server:  magicdns.localhost-tailscale-daemon
Address:  fd7a:115c:a1e0::53

Non-authoritative answer:
Name:    cloud.kacpersliwka.pl
Address:  192.168.1.13
```

Działanie przez Tailscale – będąc poza domem, tunel VPN kieruje mnie prosto do domowego serwera DNS i lokalnego proxy.



Połączenie przez telefon przy użyciu danych komórkowych, nie Wi-Fi, też działa.